Graphical Abstract

Eclipse-Modulated Transient Energy Pumping in Near-Polar Low Earth Orbit for Spin-Stabilized Solar Sails

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Highlights

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- Eclipse-modulated SRP can produce bounded transient semi-major-axis gain in near-polar LEO for passive solar sails.
- Gain exists only in a narrow LTDN–pitch region; eclipse causally enables it by suppressing negative-work orbital arcs.
- The effect is self-limiting: apogee pumping, eccentricity growth, and perigee erosion drive eventual breakdown.

Eclipse-Modulated Transient Energy Pumping in Near-Polar Low Earth Orbit for Spin-Stabilized Solar Sails⋆

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ABSTRACT

This paper investigates whether a passively oriented, spin-stabilised solar sail in near-polar low Earth orbit can exhibit a bounded interval of net orbital-energy gain through eclipse-modulated solar-radiation-pressure (SRP) forcing. Using a reduced analytical model and three-dimensional numerical propagations, we show that such a regime does exist, but only within a narrow region of LTDN–pitch space. At 700 km, only 89 of 900 sampled cases ($\sim 10\%$) exhibit positive semi-major-axis excursion within the 14-day analysis window, with the strongest case reaching $\Delta a \approx +20.07$ km at LTDN 12.266 h and pitch 24.8°. Eclipse is shown to be causally decisive: for the representative gain case, the eclipse-on trajectory reaches $\Delta a \approx +10.31$ km by day 14, whereas the eclipse-off counterpart falls to $\Delta a \approx -47.75$ km. Orbit-resolved SRP work diagnostics show that gain occurs only when eclipse suppresses predominantly negative-work portions of the SRP cycle. Longer-horizon propagation shows that this effect is transient and self-limiting rather than sustained. The representative gain case reaches a peak semi-major axis of 7098.211 km at day 26.50, but declines to 7064.717 km by day 75, below its initial value of 7078.137 km. The early increase in semi-major axis is driven mainly by apogee growth rather than uniform orbit raising, together with perigee erosion and eccentricity growth, which increase drag exposure and weaken the favourable SRP projection. The paper therefore establishes a narrower result: eclipse timing can enable a real but bounded transient energy-pumping regime in near-polar LEO, but the same coupled dynamics that permit the early gain also drive its later breakdown.

1. Introduction

Solar radiation pressure (SRP) provides a continuous, propellant-free means of modifying spacecraft orbits, and has been demonstrated in missions including IKAROS (Tsuda et al., 2013; Sawada et al., 2011), LightSail 2 (Spencer et al., 2021), and the Advanced Composite Solar Sail System (Wilkie, 2021). Recent reviews summarise the current state of solar sailing technology and mission heritage (Johnson et al., 2012; Ya Kezerashvili and Ancona, 2025). While active solar sailing enables controlled trajectory shaping, the possibility of achieving meaningful orbital energy change under purely passive configurations remains less well understood, particularly in low Earth orbit (LEO) where atmospheric drag and gravitational perturbations are significant. The use of solar sails in LEO has been explored for deorbiting (Stohlman and Lappas, 2014; Lücking et al., 2012) and for altitude maintenance (Berthet and Suzuki, 2023), but the short-horizon interaction between eclipse timing and SRP work balance has received limited attention.

Prior work has examined the coupled effects of solar radiation pressure and atmospheric drag on high area-to-mass spacecraft. In particular, Colombo and McInnes (2011) demonstrated that asymmetric SRP, including the influence of Earth eclipse, can be exploited to balance drag losses and produce long-lived Earth-centred orbits without active control. The influence of SRP on high area-to-mass ratio objects has also been studied in the context of long-term orbit evolution and resonance structures (Valk et al., 2008). These analyses highlight the importance of SRP–drag interaction and shadow-induced asymmetries in shaping orbital evolution. However, much of the existing SRP–drag literature emphasizes long-term or averaged behavior and equilibrium-like manifolds or "balance" conditions, rather than isolating short-horizon regimes in which the timing of eclipse arcs alters the sign and magnitude of orbit-averaged SRP work.

In near-polar LEO, the interaction between SRP, atmospheric drag, Earth's oblateness (J_2), and eclipse timing produces a dynamically coupled environment in which small perturbations can lead to non-intuitive orbital behaviour over short timescales (Prieto et al., 2014). In particular, a passively oriented, spin-stabilised sail experiences alternating phases of positive and negative SRP work over each orbit,

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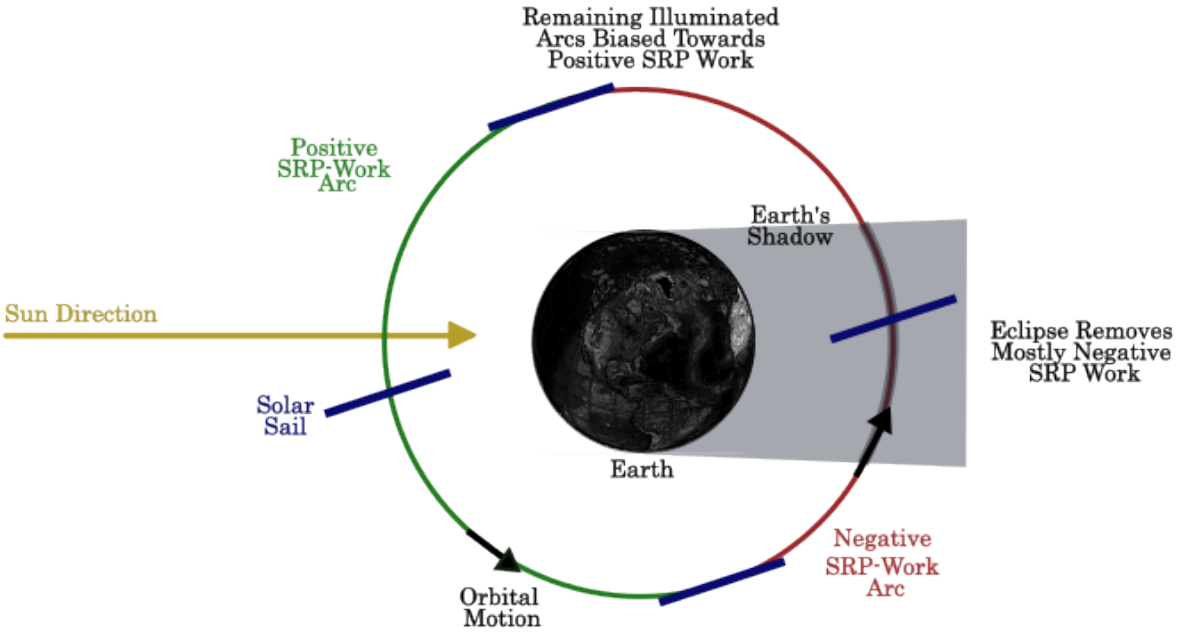


Figure 1: Conceptual schematic of the eclipse-modulated SRP work asymmetry in the reduced mechanism picture. For an inertially fixed sail, solar radiation pressure contributes positive work over part of the illuminated orbit and negative work over another part. If eclipse removes a larger fraction of the unfavourable arc than of the favourable arc, the one-orbit SRP work integral becomes positive. The figure is intended to illustrate the phase-selective gating mechanism, not to represent the full propagated 3D geometry directly.

which would be expected to largely cancel in the absence of asymmetries (McInnes, 1999).

However, eclipse introduces a temporal modulation of SRP by removing solar forcing over specific orbital arcs (Hubaux and Lemaitre, 2013). Depending on the orbital geometry, this can selectively suppress phases in which SRP performs negative work, altering the orbit-averaged energy balance. This raises the question of whether a net increase in semi-major axis can arise over a bounded time horizon under restricted conditions, without active attitude control.

This work investigates that question by isolating and characterising a transient, geometry-dependent energy-pumping regime in near-polar LEO for spin-stabilised solar sails. The aim is not to propose a mission architecture or demonstrate sustained orbit raising, but to identify where the regime exists, explain the eclipse-modulated SRP work-imbalance mechanism, and quantify the conditions under which it appears and breaks down. Using a reduced-order dynamical model and parameter-space sweeps over local time of descending node (LTDN) and sail orientation, we show that short-horizon semi-major-axis growth can occur within a narrow alignment envelope, but that the effect is inherently self-limiting through coupled eccentricity growth, increased drag exposure near perigee, and gradual drift in Sun-relative geometry.

2. Scope and Problem Definition

This work is framed as a bounded orbital-dynamics study of a specific short-horizon regime in near-polar LEO in which solar radiation pressure (SRP) can, under restricted passive conditions, produce a net orbit-averaged energy gain. The system considered is intentionally limited to the coupled effects of central gravity, Earth's oblateness (J_2), atmospheric drag, solar radiation pressure, and eclipse-modulated shadowing, for a spin-stabilised sail whose orientation is fixed in inertial space and not actively controlled.

The primary question is whether eclipse timing can, over a bounded horizon, produce a net positive orbit-averaged energy input by selectively suppressing phases of negative SRP work, and how strongly that effect depends on orbital geometry. The purpose is therefore to isolate and explain a geometric dynamical mechanism, not to demonstrate an operational orbit-raising strategy.

Several exclusions are imposed to maintain a controlled scope. This study does not claim disposal-compliant mission design, communications-feasible spacecraft architecture, collision-risk-assessed operations, maintenance of an exact Sun-synchronous orbit, or long-term sustained orbit raising. It does not attempt subsystem-level mission design, active attitude-control implementation, guidance or feedback control, or full external validation against high-fidelity industrial flight-dynamics tools. The results should therefore not be interpreted as a demonstrated mission concept, but only as evidence for a bounded dynamical effect within the modelled regime.

The objective is to map a constrained dynamical regime, explain its underlying physical mechanism, and determine the

conditions under which the observed energy gain emerges and subsequently breaks down. Sun-synchronous-like initial conditions are used only to impose a controlled and interpretable Sun-relative geometry for this analysis; they are not intended to assert that the vehicle remains on an exactly maintained operational SSO throughout the propagation.

3. Physical Mechanism of Eclipse-Modulated Energy Pumping

This section introduces a reduced analytical picture to isolate the core geometric mechanism by which eclipse timing can bias solar-radiation-pressure (SRP) work over an orbit. Its purpose is not to derive a design-optimal sail orientation for the full propagated problem, nor to replace the later 3D simulations, but to establish the sign logic of phase-selective forcing in the simplest setting where it can be seen clearly. The full numerical problem considered later includes three-dimensional orbital geometry, J_2-driven nodal precession, drag, eclipse-modulated illumination, and coupled evolution of the orbital elements; the reduced model is therefore used here only as a mechanistic scaffold for interpreting those results.

The reduced argument developed below is used to identify the sign structure of the SRP work integral under eclipse-modulated illumination, and thereby to motivate the later propagated results.

3.1. Orbital energy change and transverse work

The central quantity of interest is the sign of the orbit-averaged transverse SRP work, because the Gauss variational equation for semi-major axis (Vallado, 2013; Battin, 1999) directly links da/dt to the radial and transverse components of the perturbing acceleration. Writing the perturbing acceleration in the local orbital frame as radial (F_R), transverse (F_T), and normal (F_N) components, the semi-major-axis equation is

$$\frac{da}{dt} = \frac{2a^2}{h}\left[e\sin\theta\, F_R + (1 + e\cos\theta)\, F_T\right], \tag{1}$$

where F_R and F_T are the radial and transverse components of the perturbing acceleration, θ is the true anomaly, a is the semi-major axis, and h is the specific angular momentum.

For near-circular low Earth orbits ($e \approx 0$), this reduces to

$$\frac{da}{dt} \approx \frac{2a^2}{h} F_T = \frac{2}{n} F_T, \tag{2}$$

where $n = \sqrt{\mu/a^3}$ is the mean motion. The semi-major axis evolution is therefore governed primarily by the transverse component of the perturbing force. Equivalently, since the specific orbital energy is

$$\varepsilon = -\frac{\mu}{2a}, \tag{3}$$

its rate of change is

$$\dot{\varepsilon} = \mathbf{a}_p \cdot \mathbf{v} = F_T v, \tag{4}$$

for a circular orbit, where $\mathbf{a}_p$ is the perturbing acceleration and $v = \sqrt{\mu/a}$ is the orbital speed. Hence, in the near-circular setting used here, positive transverse work corresponds to increasing orbital energy and increasing semi-major axis, while negative transverse work has the opposite effect.

This makes the sign of the orbit-averaged transverse work the central quantity of interest in the reduced mechanism. In particular, if the SRP contribution satisfies

$$\int_0^T F_T\, dt > 0, \tag{5}$$

over one orbital period T, then the orbit-integrated SRP work is positive and the corresponding one-orbit change in orbital energy, and hence in semi-major axis, is favourable.

3.2. Passive symmetry and cancellation in the fully illuminated case

Consider first an idealised 2D planar circular orbit with the Sun fixed in inertial space. Let the Sun direction define the $+x$ axis, and parameterise the spacecraft position and velocity by

$$\mathbf{r} = r(\cos\theta, \sin\theta), \qquad \mathbf{v} = v(-\sin\theta, \cos\theta), \tag{6}$$

where θ is the orbital phase measured from the sunward direction. For a passively oriented, spin-stabilised sail with fixed inertial attitude, let the sail normal be

$$\mathbf{n} = (\cos\beta, \sin\beta), \tag{7}$$

where β is the constant pitch angle measured from the Sun line.

For an ideal reflecting sail under constant illumination, the SRP acceleration has fixed inertial direction and may be written in the form (McInnes, 1999)

$$\mathbf{a}_{\mathrm{SRP}} = a_0 \cos^2\beta\, \mathbf{n}, \tag{8}$$

where a_0 is the characteristic SRP acceleration scale. The instantaneous specific power is then

$$\dot{\varepsilon} = \mathbf{a}_{\mathrm{SRP}} \cdot \mathbf{v} = a_0 \cos^2\beta\, \mathbf{n} \cdot \mathbf{v}. \tag{9}$$

Substituting for $\mathbf{n}$ and $\mathbf{v}$ gives

$$\mathbf{n} \cdot \mathbf{v} = v(-\sin\theta\cos\beta + \cos\theta\sin\beta) = v\sin(\beta - \theta), \tag{10}$$

so that

$$\dot{\varepsilon} = a_0 v \cos^2\beta\, \sin(\beta - \theta). \tag{11}$$

Over a fully illuminated circular orbit, using $dt = (r/v)\, d\theta$, the net specific energy change is

$$\Delta\varepsilon_{\mathrm{full}} = \int_0^{2\pi} \dot{\varepsilon}\, dt = a_0 r \cos^2\beta \int_0^{2\pi} \sin(\beta - \theta)\, d\theta. \tag{12}$$

Since

$$\int_0^{2\pi} \sin(\beta - \theta)\, d\theta = 0, \tag{13}$$

it follows that

$$\Delta\varepsilon_{\text{full}} = 0. \tag{14}$$

Equation (14) formalises the passive symmetry argument: under continuous illumination, a fixed-attitude sail experiences alternating orbital phases of positive and negative transverse work that cancel exactly over one revolution. In this idealised baseline case, the one-orbit net change in orbital energy vanishes despite non-zero instantaneous SRP forcing.

3.3. Eclipse as a selective gating mechanism

The previous cancellation result changes fundamentally once eclipse is introduced. In a midnight orbit under a cylindrical shadow model, the eclipse is centred on the anti-solar direction $\theta = \pi$ and spans the interval

$$\theta \in [\pi - \delta, \pi + \delta], \qquad \delta = \arcsin\left(\frac{R_E}{r}\right), \tag{15}$$

where R_E is the Earth radius and r is the orbital radius. The illuminated arcs are therefore

$$[0, \pi - \delta] \cup [\pi + \delta, 2\pi]. \tag{16}$$

The net specific energy change over the illuminated portion of the orbit becomes

$$\Delta\varepsilon_{\text{lit}} = a_0 r \cos^2\beta \left[\int_0^{\pi-\delta} \sin(\beta - \theta)\, d\theta + \int_{\pi+\delta}^{2\pi} \sin(\beta - \theta)\, d\theta\right] \tag{17}$$

Using the full-orbit cancellation result, this may be written more compactly as minus the contribution from the shadowed arc,

$$\Delta\varepsilon_{\text{lit}} = -a_0 r \cos^2\beta \int_{\pi-\delta}^{\pi+\delta} \sin(\beta - \theta)\, d\theta. \tag{18}$$

Evaluating the integral gives

$$\int_{\pi-\delta}^{\pi+\delta} \sin(\beta - \theta)\, d\theta = 2\sin\beta\, \sin\delta, \tag{19}$$

and hence

$$\Delta\varepsilon_{\text{lit}} = -2a_0 r \cos^2\beta\, \sin\beta\, \sin\delta. \tag{20}$$

Equation (20) shows explicitly that eclipse does not merely reduce the SRP magnitude; it breaks the passive cancellation by removing a finite interval of orbital phase from the work integral. The sign of the net energy change is therefore controlled by the sign of $-\sin\beta$. In this 2D midnight-orbit idealisation,

$$\Delta\varepsilon_{\text{lit}} > 0 \qquad \Longleftrightarrow \qquad \sin\beta < 0. \tag{21}$$

A finite branch of pitch angles therefore exists for which eclipse suppresses unfavourable work more strongly than favourable work, leaving a positive residual contribution to the one-orbit SRP work balance and hence a positive one-orbit energy change.

Within this reduced 2D midnight-orbit model, the magnitude of the one-orbit energy gain is maximised by maximising $-\sin\beta\cos^2\beta$, which gives

$$\sin^2\beta = \frac{1}{3}, \tag{22}$$

with the positive-gain branch

$$\beta^\star = -\arcsin\left(\frac{1}{\sqrt{3}}\right) \approx -35.3°. \tag{23}$$

This result is used here only to show that the eclipse-modulated work imbalance is not marginal or accidental in the reduced geometry. It is not claimed to be a design optimum for the full propagated 3D system, whose geometry, illumination history, drag losses, and coupled element evolution are substantially richer.

3.4. Dependence on orbital geometry

The reduced mechanism is intrinsically phase-selective and therefore strongly dependent on orbital geometry. In the full problem, the local time of descending node (LTDN), the Sun-relative orientation of the orbital plane, and the fixed sail attitude together determine which parts of the orbit are illuminated, which are shadowed, and how the SRP work distribution is sampled. The sign and magnitude of the orbit-integrated SRP contribution therefore depend on whether eclipse preferentially removes unfavourable or favourable work phases.

The midnight case is analytically useful because the eclipse is centred on the anti-solar point and the resulting shadow interval is symmetric in orbital phase, allowing the effect of selective gating to be shown in closed form. Away from this limiting geometry, the illuminated and shadowed arcs shift relative to the SRP work distribution, and the balance between positive and negative transverse work changes accordingly. Net semi-major-axis growth is therefore expected only over restricted regions of the full parameter space.

Crucially, the same phase-dependent forcing that produces a favourable one-orbit SRP work imbalance need not produce shape-preserving orbit raising. Gauss' variational equations imply that the detailed phasing of F_R and F_T can simultaneously drive changes in eccentricity and related orbital geometry, while out-of-plane forcing in the full 3D problem affects inclination and node. A positive orbit-averaged SRP work contribution is therefore not equivalent to uniform

Table 1
Required inclination to maintain Sun-synchronous orbits (SSO) for circular orbits ($e = 0$) at selected altitudes.

Altitude (km)	Required Inclination (°)
600	97.92
700	98.04
800	98.17
900	98.31

upward translation of the orbit. In the propagated cases examined later, the same eclipse-biased phasing that produces transient semi-major-axis gain can coexist with eccentricity growth, altered perigee and apogee behaviour, and ultimately degradation of the gain regime.

3.5. Transient nature of the mechanism

The reduced analysis establishes only that eclipse can bias the one-orbit SRP work balance in favour of net energy gain for restricted geometries. It does not establish persistence, monotonic orbit raising, or long-horizon robustness once the full coupled dynamics are restored. The propagated 3D results therefore test whether this favourable work asymmetry survives in realistic near-polar LEO motion, and how it breaks down through coupled changes in eccentricity, drag exposure, and Sun-relative geometry; these longer-horizon effects are examined in Section 7.

4. Dynamical model and assumptions

This study considers a bounded orbital-dynamics problem in near-polar low Earth orbit (LEO) intended to isolate the short-horizon interaction between solar radiation pressure (SRP), eclipse timing, atmospheric drag, and J_2-driven nodal precession. The emphasis is on whether a sail with inertially fixed orientation can exhibit short-horizon semi-major-axis growth while remaining close to Sun-synchronous orbital geometry.

A Sun-synchronous orbit (SSO) is desirable because it maintains a predictable Sun-relative geometry while allowing the orbital plane to precess with the apparent annual motion of the Sun (Vallado, 2013). For near-circular LEO, this requires a secular right ascension of the ascending node (RAAN) precession rate of approximately

$$\frac{d\Omega}{dt} \approx 0.986^\circ/\text{day}, \tag{24}$$

which is primarily supplied by Earth's oblateness through the J_2 perturbation. The corresponding nodal precession rate is (Vallado, 2013)

$$\dot{\Omega} = -\frac{3}{2} J_2 \left(\frac{R_E}{a\left(1-e^2\right)} \right)^2 \sqrt{\frac{\mu}{a^3}} \cos i. \tag{25}$$

For circular orbits ($e = 0$) in the altitude range considered here, the inclination required to satisfy Eq. (24) is close to 98°, as summarised in Table 1.

In the present work, the sail orientation is prescribed as inertially fixed over the time horizons of interest, and no active steering law is applied during propagation. This contrasts with actively steered sails (Dachwald, 2005; Quarta et al., 2023) and with active momentum management approaches such as those developed for Solar Cruiser (Cannella et al., 2021; Shen and Caverly, 2026). The fixed-attitude assumption is deliberate: it allows the orbital consequences of a fixed force geometry to be isolated directly, without introducing control-law dependence into the interpretation of the results. At the same time, it imposes an important limitation. As the orbital plane precesses and the Sun-relative geometry evolves, the orientation of the sail with respect to both the Sun and the local orbital frame cannot remain optimal indefinitely. Any favourable energy-transfer regime must therefore be understood as geometry-bounded and transient rather than as a permanently sustained operating mode.

The evolution of the orbit under weak non-Keplerian perturbations is interpreted using the Gauss Variational Equations (GVEs) (Vallado, 2013; Battin, 1999). Writing the perturbing acceleration vector $\vec{F}$ in the local orbital frame as radial (F_R), transverse (F_T), and normal (F_N) components, the orbital-element evolution is given by

$$\frac{da}{dt} = \frac{2a^2}{h} \left[e \sin\theta \, F_R + (1 + e\cos\theta) \, F_T \right], \tag{26}$$

$$\frac{de}{dt} = \frac{\mu}{h} \left[\sin\theta \, F_R + \left(\cos\theta + \frac{e + \cos\theta}{1 + e\cos\theta} \right) F_T \right] \tag{27}$$

$$\frac{di}{dt} = \frac{r\cos u}{h} \, F_N, \tag{28}$$

$$\frac{d\Omega}{dt} = \frac{r\sin u}{h\sin i} \, F_N, \tag{29}$$

where $h = \sqrt{\mu a(1-e^2)}$ is the specific angular momentum, $r = a(1-e^2)/(1+e\cos\theta)$ is the instantaneous orbital radius, θ is the true anomaly, and $u = \omega + \theta$ is the argument of latitude. A detailed derivation of the GVEs is given by Vallado (2013).

These expressions clarify the role of each perturbing component. In near-circular LEO, semi-major-axis growth is driven primarily by the in-plane force, especially the transverse component F_T, whereas inclination i and RAAN Ω evolve

Table 2
Baseline parameters used for the representative gain-case propagation and local mechanism analysis.

Parameter	Value
Initial altitude, h_0	700 km
Initial inclination, i_0	98.17°
Initial eccentricity, e_0	1×10^{-4}
Spacecraft mass, m	10 kg
Sail area, A	100 m^2
Area-to-mass ratio, A/m	10 m^2 kg^{-1}
Reflectivity coefficient, C_r	1.7
Drag coefficient, C_D	2.2

through the normal component F_N. Since Sun-synchronicity relies on the secular J_2-driven nodal precession described above, additional out-of-plane forcing should remain small if the orbit is to stay close to the desired SSO geometry.

This creates the central geometrical tension of the present study. To obtain semi-major-axis growth, the sail must produce a favourable in-plane force with a strong transverse projection, while remaining close to SSO requires the normal component F_N to stay limited. Because the sail orientation is fixed in inertial space, the Sun-relative cone angle and the orbital-frame force decomposition drift as the orbit evolves, so any favourable geometry is expected to persist only over a bounded interval.

The force model retained for this study is intentionally simple and is chosen to isolate the mechanism of interest rather than to maximise environmental fidelity. The translational dynamics include central gravity, Earth's oblateness through the J_2 term, solar radiation pressure, eclipse shadowing, and atmospheric drag. No Earth albedo model is included.

Solar radiation pressure

The SRP model is based on photon momentum transfer to a reflective flat plate (McInnes, 1999). For an idealised square sail with cone angle α relative to the Sun line, the force magnitude follows the familiar $\cos^2\alpha$ dependence for a flat-plate sail (see McInnes, 1999, Ch. 2). In the implementation used here, the SRP acceleration is written as

$$\mathbf{a}_{\mathrm{SRP}} = \frac{P_{1\ \mathrm{AU}}\, A\, C_r}{mc} \cos^2\alpha\, \hat{\mathbf{r}}, \tag{30}$$

where $P_{1\ \mathrm{AU}}$ denotes the solar radiation flux at one astronomical unit (Kopp and Lean, 2011), A is the sail area, C_r is the reflectivity coefficient, m is the spacecraft mass, and $\hat{\mathbf{r}}$ is the unit vector from the spacecraft to the Sun. The SRP acceleration is evaluated in the inertial frame and then resolved into the orbital frame for interpretation through the GVEs.

Eclipses are modelled using a binary conical shadow geometry. The spacecraft is treated as illuminated when it lies outside the Earth's umbral cone and fully shadowed when it lies within it, so that SRP is either applied in full or suppressed entirely. This treatment is sufficient for the present purpose because the paper is concerned primarily with the orbital-phase selectivity of SRP application rather than with resolving penumbral transition details. The shadowed SRP acceleration is therefore written as

$$\mathbf{a}_{\mathrm{SRP}}^{\mathrm{shadowed}} = s \cdot \mathbf{a}_{\mathrm{SRP}}, \tag{31}$$

where

$$s = \begin{cases} 1, & \text{illuminated,} \\ 0, & \text{eclipsed.} \end{cases}$$

This eclipse treatment is central to the present paper because the mechanism of interest depends not only on the magnitude of SRP, but on which orbital arcs remain illuminated and which are removed by shadow.

Atmospheric drag

Atmospheric drag is included because it remains dynamically relevant throughout the altitude range considered, particularly once eccentricity growth lowers perigee and increases local density exposure. The atmospheric density is represented using a piecewise-exponential fit to U.S. Standard Atmosphere (1976) data (NOAA et al., 1976) based on Braeunig's tabulation (Braeunig, 2013). Within each altitude interval i, the density is evaluated as

$$\rho(h) = \rho_0^{(i)} \exp\left(-\frac{h - h_0^{(i)}}{H^{(i)}}\right), \tag{32}$$

where $\rho_0^{(i)}$, $h_0^{(i)}$, and $H^{(i)}$ are the reference density, reference altitude, and scale height for that layer.

The aerodynamic model used here is a simple projected-area drag law. Atmospheric rotational velocity is not modelled; the relative flow is taken directly from the spacecraft inertial velocity. The drag acceleration is written as

$$\mathbf{a}_{\mathrm{drag}} = -\frac{1}{2}\, \rho(h)\, v^2\, C_D\, \frac{A_{\mathrm{frontal}}}{m}\, \hat{\mathbf{v}}, \tag{33}$$

where $C_D = 2.2$ is a standard value for satellites in free-molecular flow (Vallado, 2013), m is the spacecraft mass, $v = \|\mathbf{v}\|$, and $\hat{\mathbf{v}}$ is the unit vector in the velocity direction. The frontal area A_{frontal} is computed from the projection of the sail surface onto the plane normal to the instantaneous

Table 3
Summary of the principal modelling assumptions used in the orbital propagations.

Assumption / model element	Implementation used in this study
Attitude treatment	Sail orientation prescribed as inertially fixed over the horizon of interest; no active steering law applied during propagation.
Central gravity	Included as the baseline point-mass Earth gravitational field.
J_2 perturbation	Included to capture Earth oblateness and the associated secular nodal precession required for near-Sun-synchronous behaviour.
Solar radiation pressure (SRP)	Included using a flat-plate sail model with one-sided illumination and force scaling with incidence angle through a cosine-based projection law.
Eclipse model	Included using a conical Earth shadow model that suppresses SRP during eclipse intervals.
Atmospheric drag	Included using a projected-area flat-plate drag law with $C_D = 2.2$.
Density model	Static piecewise-exponential atmospheric density fit based on U.S. Standard Atmosphere (1976) reference data.
Atmospheric co-rotation	Neglected; relative flow taken directly from the spacecraft inertial velocity.
Earth albedo	Omitted.
Orbit regime	Near-circular, near-polar LEO focused on maintaining approximate Sun-synchronous geometry.
Study objective	Controlled mechanism-isolation study of transient gain behaviour, not a full mission-feasibility or flight-operations model.

velocity vector, so that the effective drag area varies with the angle between the inertially fixed sail normal and the flow.

The drag model is intentionally simple. Its role in this paper is not to resolve high-fidelity free-molecular aerodynamics, but to capture the dominant loss mechanism opposing SRP-driven energy gain. This is especially important once the orbit departs from circularity, because drag then remains concentrated near perigee and can overtake the orbit-averaged SRP surplus even when early-time semi-major-axis growth is observed.

The assumptions adopted above are sufficient for the present purpose of mechanism isolation and regime mapping. They do not constitute a full validation of passive sail performance over long durations, nor do they resolve all structural, control, or environmental uncertainties relevant to flight operations. Those questions are outside the scope of this section and are addressed only insofar as they affect interpretation of the short-horizon transient dynamics studied in this paper.

5. Parameter Space and Metrics

Having defined the governing assumptions and force models, we now specify the parameter space and, critically, the fixed-window metric used to identify the transient gain regime. Because the central question of this paper is whether eclipse-modulated SRP can produce a *temporary* semi-major-axis increase over a bounded horizon, the primary map-based quantity is not the final change in a at the end of the run, and not the long-horizon change over 75 days, but the largest positive excursion above the initial semi-major axis attained at any time within a prescribed analysis window. The aim is therefore to determine whether such transient gain occupies a coherent region of design space, and how narrow or robust that region is.

The primary parameters varied in this study are the local time of the descending node (LTDN), the initial sail orientation, and the initial orbital altitude. LTDN determines the Sun-relative geometry of the orbit and hence the phasing between illuminated arcs, eclipsed arcs, and the SRP work pattern; the initial sail orientation controls the decomposition of SRP into radial, transverse, and normal components; and altitude alters both the drag environment and the baseline Sun-synchronous inclination required through 24. Together, these parameters define the geometry within which the transient gain regime may or may not appear.

In the baseline sweeps, the initial orbit is taken to be near-circular and Sun-synchronous, with inclination selected consistently with the chosen altitude using Table 1. For each case, the sail orientation is prescribed as inertially fixed from the initial epoch onward. Additional sensitivity studies may then be introduced around representative gain and non-gain cases by perturbing the nominal sail orientation, varying atmospheric density through a scalar multiplier, or applying small static orientation offsets to test the fragility of the identified regime.

The parameter-space exploration is intentionally staged. The first sweep is performed over LTDN and initial sail angle at fixed altitude and fixed sail loading, in order to determine whether a connected region of *transient* gain exists at all under the fixed-window metric defined below. Only after this primary map is established are additional altitude cases and uncertainty sweeps introduced.

A fixed analysis horizon is used for consistent comparison between cases. The primary map-based metric is

$$\Delta a_{\max}(T) = \max_{t\in[0,T]} \left(a(t) - a_0\right), \tag{34}$$

with $T = 14\,\mathrm{d}$ in the principal parameter-space sweep. Thus, $\Delta a_{\max,14\mathrm{d}}$ measures the largest positive semi-major-axis excursion above the initial orbit attained *anywhere* within the first 14 days. It is *not* the final value of $a(14\,\mathrm{d}) - a_0$, and it is *not* the net change after the longer 75-day propagation used later to study breakdown. This choice is deliberate: for a transient-gain problem, the physically relevant first question is whether a trajectory ever enters a favourable interval of positive semi-major-axis excursion within a common bounded window, not whether that excursion survives indefinitely.

To support physical interpretation, the fixed-window excursion metric is supplemented by orbit-resolved work diagnostics. In particular, we track the cumulative contribution of SRP and drag to the along-track work budget, since the mechanism proposed in Section 3 is not merely that SRP is present, but that eclipse can selectively retain or suppress SRP work over specific orbital arcs and thereby produce, or fail to produce, a temporary positive excursion in a. For a given perturbing acceleration $\mathbf{a}_p$, the instantaneous specific power is

$$P_p(t) = \mathbf{a}_p(t) \cdot \mathbf{v}(t), \tag{35}$$

and the associated cumulative specific work over a time interval $[t_1, t_2]$ is

$$W_p(t_1, t_2) = \int_{t_1}^{t_2} \mathbf{a}_p(t) \cdot \mathbf{v}(t)\, dt. \tag{36}$$

In practice, these quantities are evaluated separately for SRP and drag and then accumulated over individual orbits and over the full analysis horizon.

Because the hypothesised mechanism is most naturally interpreted in the orbital frame, we also track the transverse components of the perturbing accelerations. From Eq. (26), the transverse contribution is especially important in near-circular orbits, where it acts most directly on orbital energy and semi-major axis. For this reason, we record both the instantaneous transverse accelerations $F_{T,\mathrm{SRP}}$ and $F_{T,\mathrm{drag}}$, and their orbit-integrated contributions, alongside the full energy-based work budget. These diagnostics are used to distinguish between three physically different situations: cases in which eclipse removes predominantly unfavourable SRP arcs, cases in which eclipse produces little net bias, and cases in which the remaining illuminated arcs are themselves energetically unfavourable.

In addition to semi-major axis, we record the evolution of apogee radius r_a, perigee radius r_p, eccentricity e, inclination i, and right ascension of the ascending node Ω. These quantities are necessary because positive Δa alone does not imply clean whole-orbit raising. In particular, a gain in a may be accompanied by growth in eccentricity, with apogee increasing more strongly than perigee. Since drag remains concentrated near perigee, this distinction is central to understanding why the transient gain regime can disappear even after an initially favourable interval. The coupled evolution of r_a, r_p, and e therefore forms a second layer of interpretation beyond the headline Δa result.

To capture the role of eclipse geometry explicitly, we also record the eclipse state as a binary quantity throughout propagation, together with the eclipse fraction of each orbit. This allows the work budget to be partitioned orbit-by-orbit into illuminated and shadowed segments. In representative cases, these diagnostics are later plotted against true anomaly so that the reader can see directly which orbital arcs contribute positive SRP work, which contribute negative SRP work, and which are removed by eclipse. This is an important distinction: the paper does not argue merely that eclipse reduces SRP, but that eclipse can alter the sign balance of orbit-averaged SRP work.

On the basis of the fixed-horizon metric $\Delta a_{\max}(T)$, each simulation is assigned to one of three qualitative classes:

- **Gain case:** $\Delta a_{\max}(T) > 0$, indicating that the trajectory attains a positive semi-major-axis excursion at some point within the window;
- **Neutral case:** $\Delta a_{\max}(T) \approx 0$, indicating that no meaningful positive excursion is attained within the window;
- **Decay case:** cases for which $a(t) - a_0$ remains non-positive throughout the window, so that $\Delta a_{\max}(T)$ is zero or numerically negligible and the trajectory is effectively monotone non-gaining over that interval.

These labels are descriptive rather than ontological. In particular, "gain" here means *transient gain within the prescribed window*, not sustained long-horizon orbit raising; likewise, the strongest gain cell in the map is the case with the largest $\Delta a_{\max,14\mathrm{d}}$, not necessarily the most favourable long-horizon trajectory once eccentricity growth and drag accumulation are taken into account.

For representative cases drawn from each class, we additionally identify a transient gain window from the cumulative SRP and drag work curves. This secondary diagnostic is used later to distinguish early favourable behaviour from the subsequent drag-dominated breakdown phase. It is therefore interpretive rather than classificatory: the parameter-space map itself is defined only by $\Delta a_{\max,14\mathrm{d}}$.

The minimum set of diagnostics recorded for each run is therefore as follows: instantaneous SRP and drag accelerations, their transverse components, eclipse state, classical orbital elements, cumulative SRP and drag work, orbit-by-orbit eclipse fraction, and the evolution of a, r_a, and r_p. This choice is deliberate. The present paper is a mechanism and regime-mapping study, so the recorded metrics must do more than rank cases numerically: they must also explain why a given case gains, remains neutral, or decays.

The results that follow therefore proceed in three steps. First, the fixed-window metric $\Delta a_{\max,14\mathrm{d}}$ is used to identify where transient gain does and does not occur in parameter space. Second, orbit-resolved work budgets are used to test

whether eclipse causally biases the SRP work balance in those cases. Third, apogee, perigee, eccentricity, and cumulative drag/SRP comparisons are used to explain why a case that exhibits transient gain over 14 days need not remain favourable over longer horizons.

6. Existence and Mechanism of the Transient Gain Regime

This section addresses four questions in sequence: whether a bounded transient-gain region exists at 700 km, whether eclipse is causally responsible for it, how that causality appears in the orbit-resolved SRP work budget, and why the representative gain case should still be interpreted as transient rather than sustained. Because the paper is concerned with *transient* gain, the map-based metric used here is the maximum semi-major-axis increase attained within the first 14 days, not the final Δa at the end of the run.

Figure 2 presents the fixed-window map of maximum semi-major-axis increase at 700 km as a function of LTDN and initial pitch angle. A single dominant positive lobe appears over approximately LTDN 9.0 to 14.75 h and pitch 0° to 43.4°, with its strongest cell at LTDN 12.266 h and pitch 24.8°, where the trajectory reaches a maximum Δa of approximately +20.07 km. Two much smaller retrograde-pitch clusters also appear, but their gains remain below 0.14 km and are negligible relative to the main lobe. Overall, only 89 of 900 sampled cells ($\sim$ 10%) exhibit positive transient SMA increase, so non-gain behaviour dominates most of the map.

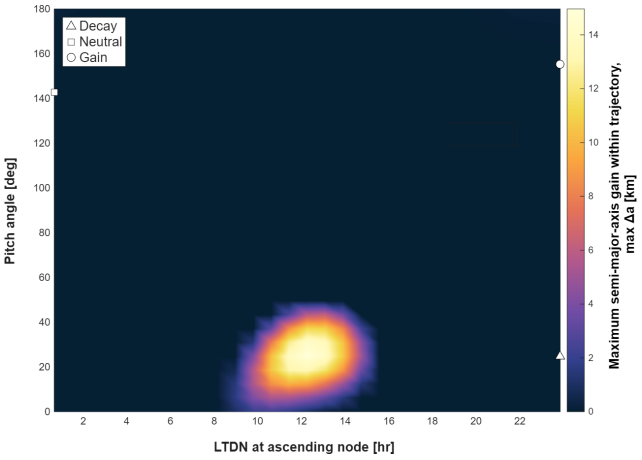


Figure 2: Maximum semi-major axis change (Δa) at 700 km as a function of LTDN and sail pitch angle, showing a localized region of net orbital energy gain with dominant decay outcome in the surrounding space.

The restricted extent of the positive region indicates that the effect is geometry-selective rather than a generic passive orbit-raising tendency.

Three representative 700 km cases are used throughout the remainder of the section: a gain case at RAAN 285.5°, LTDN 12.27 h, and pitch 24.8°; a neutral case at RAAN 111.7°, LTDN 0.68 h, and pitch 142.8°; and a decay case at RAAN 99.3°, LTDN 23.85 h, and pitch 155.2°. The gain case reaches a maximum $\Delta a \approx +20.07$ km at approximately day 26.5, but later declines to $\Delta a \approx -13.42$ km by day 75; the decay case loses SMA monotonically and re-enters by day 42.8. These cases are used to separate bounded existence from mechanism.

The most direct causal test is to compare the representative gain case with and without eclipse while holding all other initial conditions and model assumptions fixed. Figure 3 shows that the short-horizon outcome reverses under that single change. By day 14, the eclipse-on case reaches $\Delta a \approx +10.31$ km, whereas the eclipse-off counterpart has fallen to $\Delta a \approx -47.75$ km, a separation of approximately 58.06 km. Over the full 75-day propagation, the eclipse-on case survives to the end of the run with final $\Delta a \approx -13.42$ km, whereas the eclipse-off case re-enters. The gain case is therefore not simply SRP-favourable in a generic sense; its transient improvement depends on eclipse.

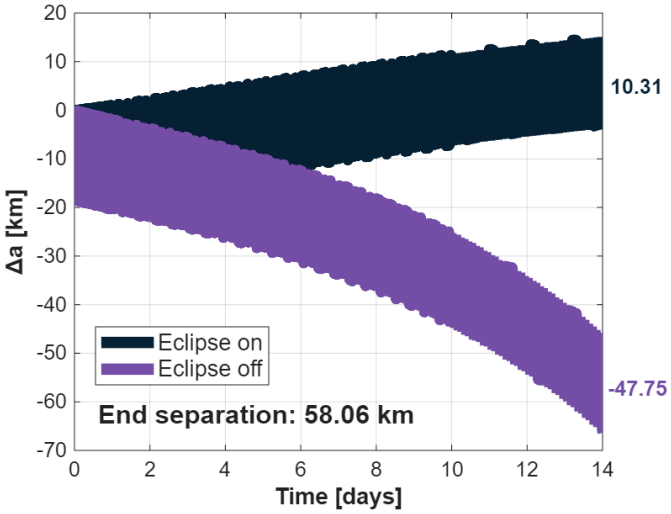


Figure 3: Comparison of semi-major-axis evolution with eclipse on and off for the representative 700 km gain case, showing that the favourable transient behaviour depends directly on eclipse: with eclipse on, the case reaches $\Delta a \approx +10.31$ km by day 14, whereas the eclipse-off counterpart has already fallen to $\Delta a \approx -47.75$ km.

The eclipse-on/off comparison establishes causality, but not the orbit-scale mechanism. Figure 4 addresses that mechanism by plotting the instantaneous SRP specific power, $\mathbf{a}_{\mathrm{SRP}} \cdot \mathbf{v}$, over one periapsis-to-periapsis orbit for the representative gain, neutral, and decay cases, together with the eclipse interval and a dashed no-eclipse reference. In the gain case, eclipse falls predominantly on a would-be negative-power arc: 85.7% of eclipsed samples correspond to negative no-eclipse SRP power, and by integrated magnitude eclipse suppresses roughly $-860\,\mathrm{J\,kg^{-1}}$ of harmful SRP work while removing only about $+15\,\mathrm{J\,kg^{-1}}$ of beneficial

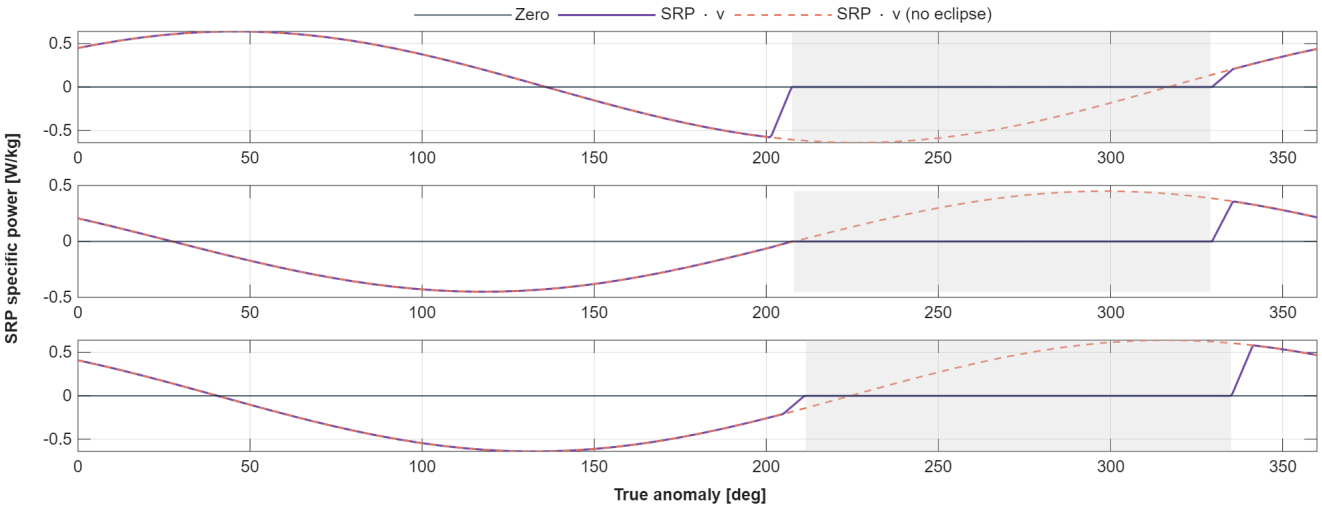


Figure 4: Instantaneous SRP specific power, $\mathbf{a}_{\mathrm{SRP}} \cdot \mathbf{v}$, over one periapsis-to-periapsis orbit for the representative gain, neutral, and decay cases. The shaded interval marks eclipse, and the dashed curve shows the no-eclipse reference. In the gain case, eclipse suppresses a predominantly negative-power arc; in the neutral and decay cases, it instead suppresses a predominantly positive-power arc.

work. In the neutral and decay cases, the sign structure reverses: 95.2% and 85.7% of eclipsed samples respectively correspond to would-be *positive* SRP power, so eclipse suppresses the recovery arc instead. The mechanism is therefore sign-selective: the outcome depends on which branch of the SRP-power cycle is removed by shadow.

Figure 4 therefore identifies the mechanism at the instantaneous level: eclipse changes the sign of the orbit-integrated SRP contribution by removing different branches of the SRP-power cycle in different geometries.

Figure 5 shows the same effect in cumulative form. In the gain case, the eclipse-on orbit closes with cumulative SRP specific work of $+799.29\,\mathrm{J\,kg^{-1}}$, compared with $-9.69\,\mathrm{J\,kg^{-1}}$ for the no-eclipse reference. In the neutral and decay cases, the eclipse-on end values are instead $-665.25\,\mathrm{J\,kg^{-1}}$ and $-857.93\,\mathrm{J\,kg^{-1}}$, while the corresponding no-eclipse references end near zero at $-4.81\,\mathrm{J\,kg^{-1}}$ and $-53.90\,\mathrm{J\,kg^{-1}}$. The cumulative view therefore confirms the sign-selective interpretation inferred from Figure 4: eclipse enhances the one-orbit SRP balance only in the gain geometry and degrades it in the neutral and decay geometries.

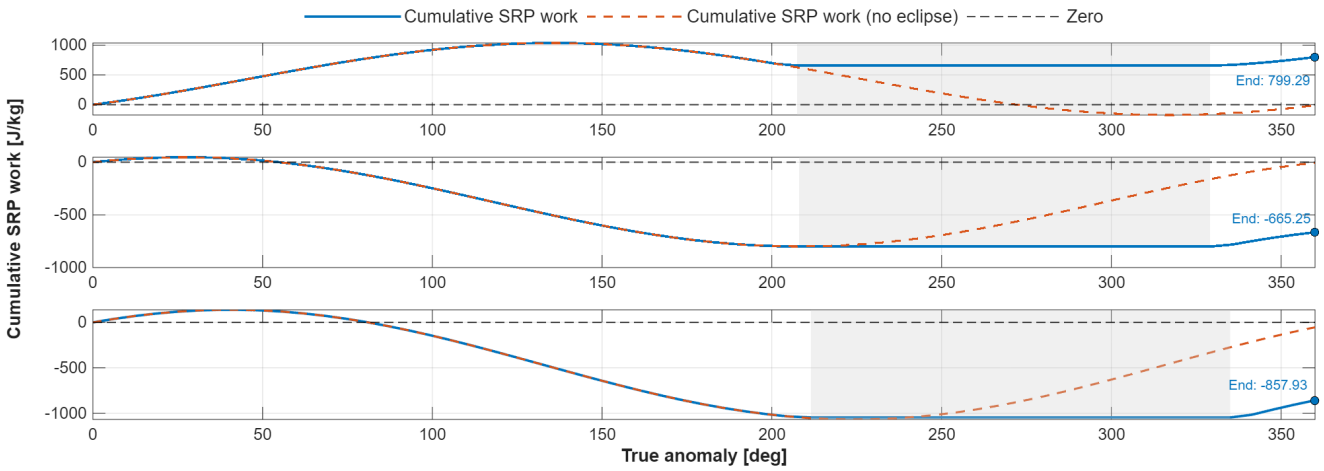


Figure 5: Cumulative SRP specific work over one periapsis-to-periapsis orbit for the representative gain, neutral, and decay cases, with dashed no-eclipse references for comparison. The eclipse-on end values are $+799.29\,\mathrm{J\,kg^{-1}}$, $-665.25\,\mathrm{J\,kg^{-1}}$, and $-857.93\,\mathrm{J\,kg^{-1}}$ respectively, showing that eclipse enhances the SRP budget only in the gain geometry and worsens it in the neutral and decay geometries.

The representative gain case also confirms an important limitation of the mechanism. Although the trajectory attains a substantial positive semi-major-axis excursion within the analysed window, that behaviour is not equivalent to clean or sustained whole-orbit raising over longer horizons. The detailed longer-horizon evolution of the gain case is examined in Section 7.

Taken together, the results in this section establish three points. First, a bounded transient-gain region exists at 700 km in LTDN–pitch space. Second, eclipse is causally decisive: removing it reverses the short-horizon outcome of the representative gain case. Third, the orbit-resolved SRP budget identifies the mechanism: eclipse helps only when it removes a negative-power branch and harms when it removes a positive-recovery branch. The representative gain case is therefore physically real, but not a claim of sustained orbit raising.

7. Coupled Orbital Evolution and Breakdown of the Gain Regime

The representative gain trajectory must now be examined over longer horizons to determine what the early increase in semi-major axis actually corresponds to dynamically, and why that favourable behaviour does not persist. The central result is linear: the early increase in a is real, but it is produced mainly by apogee pumping rather than by a clean upward translation of the whole orbit; this increases eccentricity, lowers perigee, concentrates drag loss near perigee, and eventually causes the late-window orbit-averaged energy balance to turn negative.

Figure 6 makes the first part of that breakdown visible directly. For the representative 700 km gain case introduced in Section 6, the 75-day histories of a, apogee altitude, perigee altitude, and eccentricity show that the orbit is not lifted uniformly. Instead, apogee grows much more strongly than perigee is preserved, so the early gain develops together with a progressive departure from circularity.

The semi-major axis begins at $a_0 = 7{,}078.137\,\mathrm{km}$, rises early, reaches a maximum of 7,098.211 km, and then declines for the remainder of the propagation to a final value below its initial level. The behaviour is therefore explicitly non-monotonic: there is a finite interval of positive excursion, but not sustained whole-horizon orbit raising. In other words, the short-horizon gain is real, yet it does not remain self-maintaining over the full 75-day window.

The apogee–perigee split shows how that breakdown develops. Apogee rises strongly and remains elevated, whereas perigee falls substantially and remains depressed. At the same time, the eccentricity grows from an initially near-circular value to an order 10^{-2} state and relaxes only partially thereafter. The key point is therefore geometric rather than merely descriptive: the early increase in a is achieved primarily by stretching the orbit, not by lifting it uniformly.

The physical consequence is that drag loss becomes progressively concentrated near perigee, where the atmosphere is densest. This is the clearest breakdown pathway visible in the long-horizon outputs: the same evolution that produces the early gain also creates a lower-perigee, more eccentric orbit that is increasingly vulnerable to atmospheric loss. No stronger claim than that is needed here.

The cumulative work budget in Figure 7 must be read carefully. Over the full 75-day window, cumulative SRP work remains slightly larger than cumulative drag work, so the cumulative total is still marginally positive by day 75. That does *not* mean the regime is still strengthening at late times. It means only that an earlier accumulated surplus has not yet

Table 4
Long-horizon summary of the representative 700 km gain case over the 75-day propagation. The case corresponds to altitude 700 km, RAAN 285.5°, pitch 24.8°, LTDN 12.266 hr, inclination 98.2°, $e_0 = 10^{-4}$, sail area 100 m^2, mass 10 kg, $C_r = 1.7$, and $C_d = 2.2$.

Quantity	Initial	Peak / Minimum	Final
Semi-major axis, a [km]	7078.137	7098.211 (max, day 26.50)	7064.717
Net semi-major-axis change, $\Delta a_{75\mathrm{d}}$ [km]	−13.420		
Apogee altitude [km]	700.709	864.548 (max, day 39.50)	775.102
Perigee altitude [km]	699.293	546.370 (min, day 42.17)	598.059
Eccentricity, e	1.000×10^{-4}	2.176×10^{-2} (max, day 40.72)	1.253×10^{-2}
Cumulative SRP work at day 75 [MJ]	+6.054		
Cumulative drag work at day 75 [MJ]	−6.030		
Cumulative total work at day 75 [MJ]	+0.024		
Work ratio at day 75, $\lvert W_{\mathrm{drag}}\rvert / \lvert W_{\mathrm{SRP}}\rvert$	0.9961		

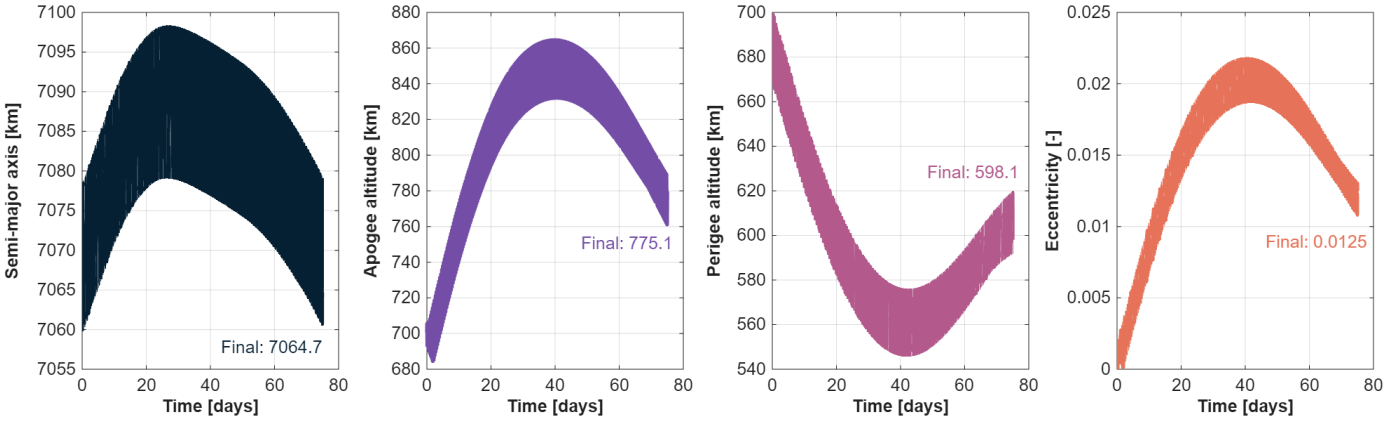


Figure 6: Long-horizon evolution of orbital elements for the representative 700 km gain case, showing non-monotonic semi-major-axis evolution, strong apogee growth, perigee erosion, and eccentricity increase over the 75-day propagation.

been fully erased. By the end of the run, that surplus is nearly exhausted, with $|W_{\mathrm{drag}}|/|W_{\mathrm{SRP}}| \approx 0.9961$.

The more revealing diagnostic is therefore the orbit-averaged early-versus-late comparison in Figure 8. Early in the run, the mean SRP contribution exceeds the mean drag loss, so the per-orbit balance is positive. Late in the run, the mean SRP contribution has collapsed while drag remains large, and the orbit-averaged net balance becomes negative. The breakdown is therefore already present on a local per-orbit basis even though the full-window cumulative total remains slightly positive.

The correct long-horizon interpretation is therefore a timescale split: the trajectory retains a small cumulative surplus from its earlier favourable phase, but its late-time orbit-averaged balance has already turned negative. The regime is not sustained; it is being unwound.

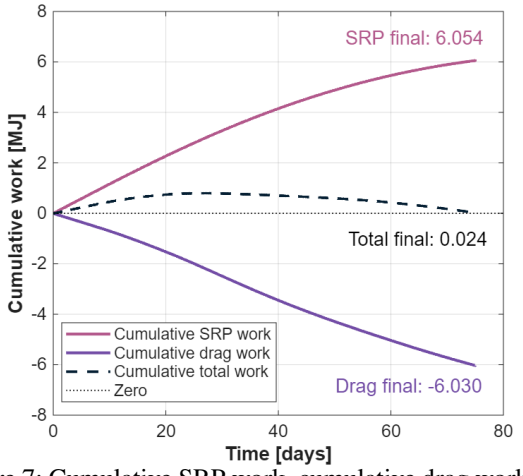


Figure 7: Cumulative SRP work, cumulative drag work, and cumulative total work for the representative 700 km gain case over 75 days. The cumulative budget remains positive throughout the analysed window, but only narrowly: by day 75, cumulative drag has nearly caught up to cumulative SRP.

The physical picture is therefore one of temporary outperformance, not eliminated decay. Eclipse-modulated SRP produces an early advantage, but the resulting orbital distortion

Table 5
Early-versus-late comparison for the representative 700 km gain case. The orbit-averaged work budget uses 20% windows (first and last 218 of 1090 total orbits), whereas the SRP force-decomposition comparison uses 10% windows (first and last 109 orbits), matching the conventions used in the corresponding figure-generation scripts.

Metric	Early window	Late window	Late / Early
Orbit-averaged work budget (20% windows; 218 orbits each)			
Mean SRP work [kJ/orbit]	+7.91	+2.71	0.343
Mean drag work [kJ/orbit]	−5.01	−4.55	0.908
Mean net work [kJ/orbit]	+2.89	−1.84	—
SRP force decomposition (10% windows; 109 orbits each)			
Orbit-mean $\lvert F_T \rvert$ [N]	3.52×10^{-4}	5.91×10^{-5}	0.207
Orbit-mean $\lvert F_N \rvert$ [N]	5.89×10^{-7}	1.50×10^{-4}	5.36

lowers perigee, increases eccentricity, and makes subsequent drag losses harder to offset. The mechanism delays deterioration for a finite interval; it does not remove the long-horizon tendency toward decay.

The longer-term weakening is also consistent with the force-decomposition framework introduced in Chapter 4. There, using the Gauss Variational Equations, we argued that semi-major-axis growth depends primarily on maintaining a favourable in-plane force, especially a strong transverse component F_T, while avoiding excessive projection into the normal direction F_N, which does not contribute directly to useful in-plane energy growth. The representative gain case shows a major redistribution of the SRP force between these channels. As shown in Figure 9, the orbit-mean magnitude of the transverse component decreases from approximately 3.529×10^{-4} N early in the mission to approximately 7.312×10^{-5} N late in the mission, a reduction by a factor of about 4.8. Over the same early-to-late comparison, the orbit-mean magnitude of the normal component increases from approximately 3.224×10^{-5} N to approximately 1.727×10^{-4} N, an increase by a factor of about 5.4.

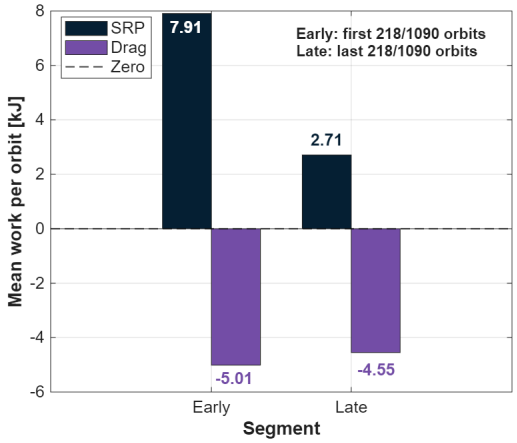


Figure 8: Early-versus-late orbit-averaged work budget for the representative 700 km gain case. The mean SRP contribution collapses strongly between the early and late windows, while the drag penalty remains large, producing a negative late-window net balance.

This redistribution of SRP force is not secondary bookkeeping. Because the sail orientation is fixed in inertial space while the orbit evolves, the SRP force becomes progressively less well aligned with the favourable local RT-plane direction. The transverse component weakens strongly, the normal component becomes relatively larger on an early-to-late basis, and the contribution to $\dot{a}$ correspondingly degrades. This geometrical drift is therefore a second breakdown pathway, alongside the drag concentration caused by perigee erosion.

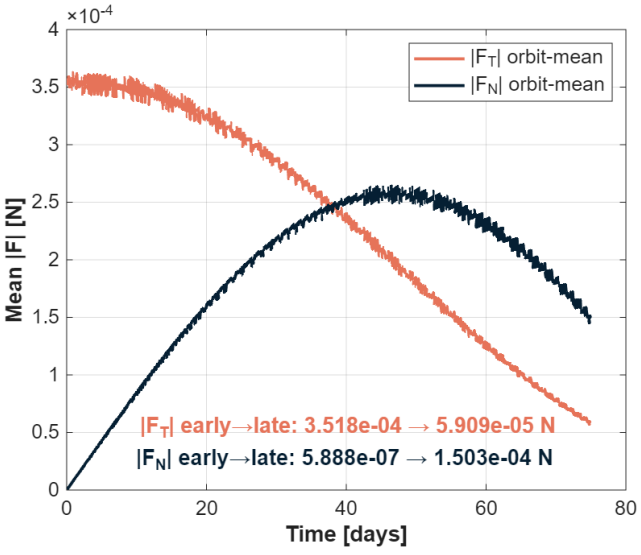


Figure 9: Orbit-mean SRP force decomposition for the representative 700 km gain case. The favourable transverse component weakens strongly over time, while the normal component becomes much larger on an early-to-late basis, indicating a substantial redistribution away from the most useful in-plane forcing direction.

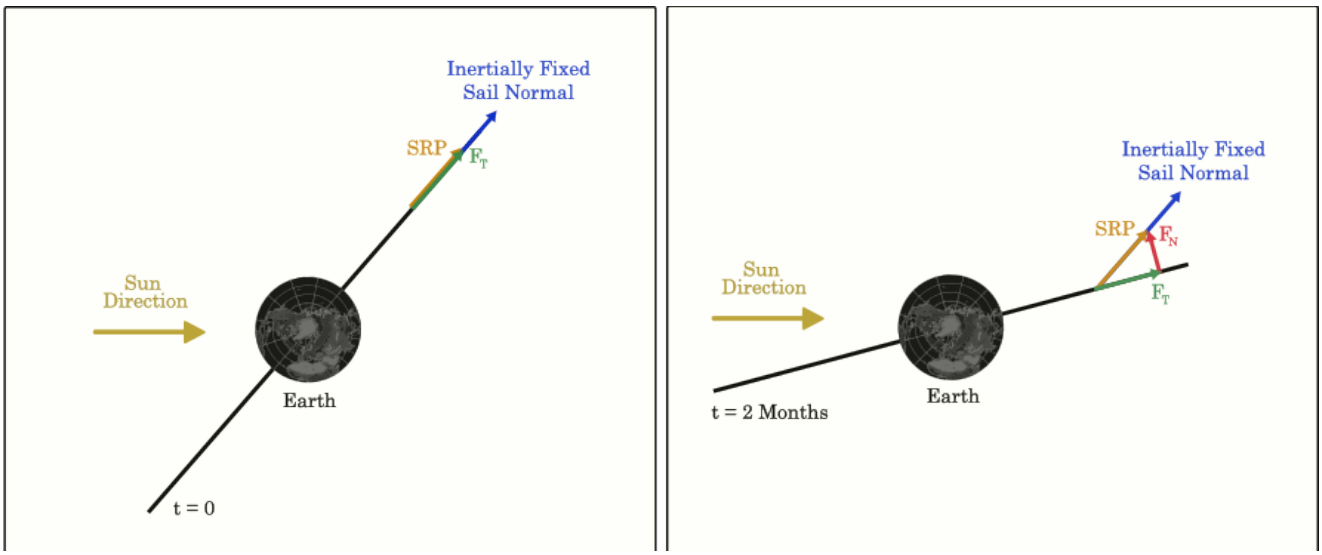


Figure 10: Illustration of SRP projection drift for an inertially fixed, spin-stabilised sail in a precessing near-polar orbit. At $t = 0$ (left), the fixed sail normal produces a favourable transverse SRP component F_T. After two months (right), orbital precession changes the Sun-relative orbital geometry, so the same inertially fixed attitude projects less favourably into the local frame, with reduced F_T and larger normal contribution F_N. This geometric drift contributes to the breakdown of the transient gain regime over longer timescales.

Taken together, the results in this section resolve the apparent tension between early gain and later decay. The mechanism established in Section 6 is real, but it is front-loaded and self-limiting. Early positive Δa is achieved mainly through apogee pumping rather than clean whole-orbit lifting; that evolution increases eccentricity and lowers perigee, which concentrates drag loss, while the fixed inertial sail orientation also causes the SRP force to drift away from the most favourable transverse direction. The late-time orbit-averaged balance therefore turns negative even though a small cumulative surplus from the earlier phase remains by day 75. The regime does not fail abruptly; it progressively exhausts the margin that created it.

8. Robustness and sensitivity of the gain-anchor case

To assess whether the identified short-horizon gain behavior is a numerically isolated point or instead occupies a finite bounded neighborhood, a set of local sensitivity sweeps was carried out about the same representative gain-anchor geometry over a fixed 14-day analysis window. In all cases, the response metric is the maximum semi-major-axis increase attained within that window,

$$\Delta a_{\max,14\mathrm{d}} = \max_{t\in[0,14\,\mathrm{d}]} \left(a(t) - a_0\right), \tag{37}$$

expressed in kilometres. This is therefore not the final semi-major-axis change at day 14, but the largest positive excursion relative to the initial orbit attained at any time within the first 14 days. The common reference geometry for all three sweeps is defined by $\Omega = 285.5°$, LTDN $= 12.266176\,\mathrm{hr}$, nominal pitch 24.8°, zero roll offset, a $10 \times 10\,\mathrm{m}$ sail, and $A/m = 10\,\mathrm{m^2\,kg^{-1}}$, with the same configuration propagated at both 700 km and 800 km. The corresponding nominal responses are $\Delta a_{\max,14\mathrm{d}} = 14.958\,\mathrm{km}$ at 700 km and 33.937 km at 800 km. The 800 km curves should therefore be interpreted as a same-geometry altitude comparison rather than as a separately re-optimised gain anchor.

8.1. Static roll sensitivity

At the same time, the envelope remains clearly bounded at both altitudes. As the roll magnitude increases beyond roughly 20°, the attainable gain decreases steadily, showing that the eclipse-biased work surplus is still alignment-sensitive. The important additional result is that the sensitivity weakens markedly at higher altitude. At 700 km, offsets of ±45° retain only a small fraction of the zero-roll peak, whereas at 800 km the same offsets still preserve roughly half of the nominal response. In absolute terms, the 800 km case with a 45° roll offset still produces a gain comparable to the zero-roll nominal case at 700 km. The sweep remains mildly asymmetric about zero, consistent with the present one-sided SRP illumination model, but the dominant result is not the asymmetry itself; it is that the admissible roll envelope broadens substantially once the drag environment is relaxed.

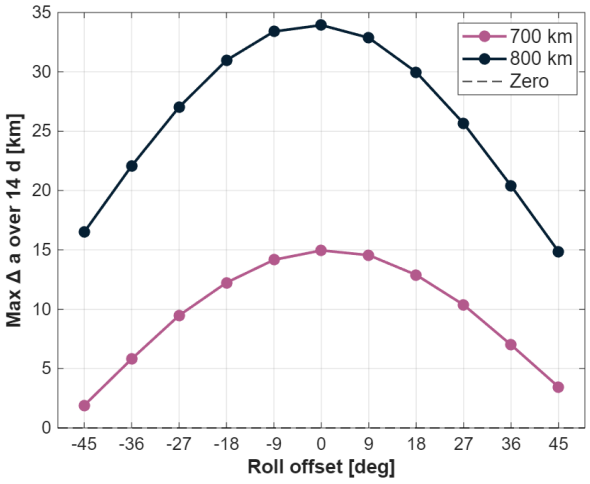


Figure 11: Sensitivity of the 14-day maximum semi-major-axis increase, $\Delta a_{\mathrm{max,14d}}$, to static roll offset about the gain-anchor attitude, shown for the same reference geometry propagated at 700 km and 800 km. The common geometry is $\Omega = 285.5°$, LTDN = 12.266176 hr, pitch 24.8°, zero roll offset, and $A/m = 10\,\mathrm{m^2\,kg^{-1}}$. The response peaks at zero roll at both altitudes, but the bounded alignment envelope is substantially broader at 800 km.

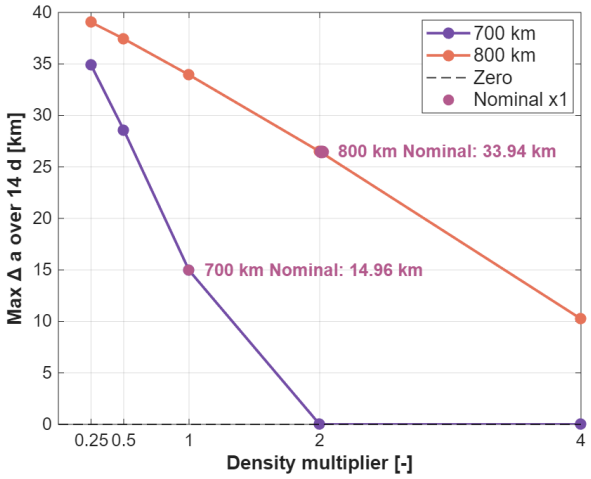


Figure 12: Sensitivity of the 14-day maximum semi-major-axis increase, $\Delta a_{\mathrm{max,14d}}$, to a multiplicative density scaling applied about the same gain-anchor geometry at 700 km and 800 km. Reduced density strengthens the short-horizon gain at both altitudes. Increased density suppresses the 700 km response completely by 2× nominal density, whereas the 800 km case remains positive throughout the tested range, showing that the drag-limited boundary shifts substantially with altitude.

8.2. Sensitivity to atmospheric density level

Figure 12 shows the effect of scaling the atmospheric density model by a constant multiplicative factor while holding the gain-anchor geometry fixed at both 700 km and 800 km. At 700 km, the nominal atmosphere case (1×) gives $\Delta a_{\mathrm{max,14d}} = 14.96$ km, which increases to 28.59 km at 0.5× density and 34.89 km at 0.25×. Increasing density, however, suppresses the gain completely at this altitude: at 2× and 4× density, $\Delta a_{\mathrm{max,14d}} = 0$. At 800 km, by contrast, the same sweep remains positive throughout the full tested range, with 33.94 km at nominal density, 26.49 km at 2×, and 10.25 km even at 4× density.

This sweep shows most clearly that drag is the dominant bounding mechanism, but it also shows that the location of the drag-limited boundary depends strongly on altitude. At 700 km, the short-horizon semi-major-axis increase exists only while the eclipse-biased SRP work surplus exceeds the accumulated drag losses; once the atmosphere is strengthened sufficiently, the drag penalty dominates from the outset and the semi-major axis never rises above its initial value within the 14-day window. At 800 km, however, the same mechanism remains much more resilient: increased density degrades the gain substantially, but does not eliminate it across the tested range. The most defensible interpretation is therefore not that the regime is uniformly fragile, but that it is strongly altitude-dependent and can lie either near or well away from the drag-limited boundary depending on orbital height.

8.3. Sensitivity to area-to-mass ratio

Figure 13 shows the response to varying area-to-mass ratio while keeping the spacecraft mass fixed at 10 kg. The square sail side length is varied from 5 to 15 m, corresponding to A/m values from 2.5 to 22.5 $\mathrm{m^2\,kg^{-1}}$. At 700 km, the response is non-monotonic. Starting from low A/m, the maximum short-horizon gain rises from 4.17 km at $A/m = 2.5\,\mathrm{m^2\,kg^{-1}}$ to the nominal 14.96 km at $A/m = 10\,\mathrm{m^2\,kg^{-1}}$, then reaches a broad maximum of 17.92 km at $A/m = 14.4\,\mathrm{m^2\,kg^{-1}}$, before declining slightly at larger sail sizes. At 800 km, by contrast, the response remains monotonic over the full sampled range, rising from 8.54 km at $A/m = 2.5\,\mathrm{m^2\,kg^{-1}}$ to 71.61 km at $A/m = 22.5\,\mathrm{m^2\,kg^{-1}}$, with no roll-over yet visible.

The altitude split again changes the interpretation. At 700 km, the non-monotonicity shows that increasing sail size does not produce indefinite improvement: larger A/m strengthens SRP, but it also strengthens drag and modifies the coupled orbital response over the 14-day window, producing a finite optimum and then a shallow plateau. At 800 km, however, the same drag-induced roll-over is pushed beyond the sampled range, so increasing A/m continues to improve the short-horizon gain throughout the tested envelope. The shared conclusion is that stronger SRP authority is beneficial, but the accessible benefit depends on how strongly drag competes over the observation window. Compared with the density sweep, the A/m response remains broad at both altitudes, but only the lower-altitude case shows a clear local optimum within the sampled range.

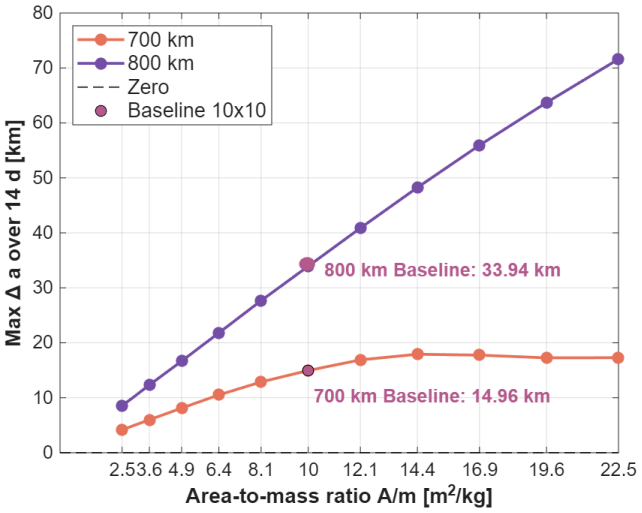


Figure 13: Sensitivity of the 14-day maximum semi-major-axis increase, $\Delta a_{\max,14d}$, to sail area-to-mass ratio for the gain-anchor case, with spacecraft mass held fixed at 10 kg, shown for the same reference geometry at 700 km and 800 km. The 700 km response is non-monotonic, with a broad optimum near $A/m = 14.4\,\mathrm{m^2\,kg^{-1}}$, whereas the 800 km response continues to increase across the full sampled range.

8.4. Interpretation of the sensitivity envelope

Taken together, the three sweeps support a restrained but more nuanced conclusion. The transient gain behavior is not a numerically singular point: it persists over a finite neighborhood in roll and over a broad range of positive A/m, and this admissible neighborhood expands significantly between 700 km and 800 km. At the same time, the mechanism is still not a generally robust passive orbit-raising mode. Its existence remains contingent on favorable Sun-orbit-attitude phasing and on a drag environment weak enough for the eclipse-biased SRP work surplus to remain positive over the observation window.

The most defensible statement is therefore that the gain occupies a bounded local envelope whose size depends materially on altitude. Roll sensitivity demonstrates finite but limited alignment tolerance at 700 km, but a substantially wider tolerance band at 800 km. Density sensitivity shows that drag remains the dominant bounding mechanism, while also showing that the regime can lie either close to or well away from the drag-dominated breakdown threshold depending on orbital height. Area-to-mass sensitivity shows that stronger SRP authority improves the short-horizon gain broadly, but that the onset of diminishing returns is itself altitude-dependent and is shifted to larger A/m at higher altitude. These results are consistent with the broader thesis of the paper: the mechanism is real and physically interpretable, but it is geometry-bounded, alignment-sensitive, and ultimately limited by the coupled drag and orbital-distortion effects that later drive the long-horizon transition from transient gain to decay.

9. Conclusions

This paper examined whether a passively oriented, spin-stabilised solar sail in near-polar low Earth orbit can exhibit a bounded interval of net orbital-energy gain under eclipse-modulated solar-radiation-pressure forcing. The results show that such a regime does exist, but only over a narrow and well-defined region of the sampled LTAN–pitch space. The gain response is therefore not a generic property of passive sailing in LEO, but a constrained transient effect that appears only under specific Sun-relative geometries.

The results also show that eclipse is the causal enabler of this transient gain regime. In the representative gain case, the inclusion of eclipse produces positive short-horizon semi-major-axis growth, whereas removing eclipse from the same initial condition eliminates that behaviour and leads instead to clear decay. Orbit-resolved SRP work diagnostics show why: eclipse does not merely weaken SRP forcing, but selectively removes unfavourable portions of the SRP cycle. Under the gain geometry, this shifts the orbit-averaged SRP work balance positive; under neutral and decaying geometries, the same shadowing instead suppresses the positive recovery branch and leaves the net SRP contribution negative. The central physical result is therefore that eclipse timing can control the sign of the orbit-averaged SRP work input.

At the same time, the longer-horizon behaviour shows unambiguously that this effect is transient and self-limiting rather than sustained. The representative gain trajectory initially exhibits semi-major-axis increase, but this does not correspond to uniform orbit raising. Instead, the orbit evolves through apogee growth, perigee erosion, and rising eccentricity, so that the early energy advantage progressively creates the conditions that undermine it. Drag becomes increasingly concentrated near perigee, while the favourable SRP projection weakens as the orbit distorts. As a result, the late-time orbit-averaged energy balance turns negative and the semi-major axis ultimately declines below its initial value. The transient gain regime identified here must therefore be understood as a front-loaded energy imbalance, not as evidence of persistent passive orbital uplift.

The sensitivity analysis supports the same interpretation. The gain-anchor case is not a numerically isolated point, but neither is it broadly robust. At lower altitude, the response is strongly limited by density and attitude offset; at higher altitude, the same geometry remains more tolerant over the sampled envelope, but still does not become a general passive orbit-raising solution. The mechanism is therefore physically real and interpretable, yet intrinsically bounded by geometry, atmospheric conditions, and the coupled eccentricity and drag penalties that emerge during the evolution.

Accordingly, the main contribution of this paper is not to propose sustained passive orbit raising in LEO, but to establish a narrower result: a constrained transient gain regime exists, eclipse timing is its causal enabler, and the same coupled dynamics that permit the early gain also drive its later breakdown. Future work should remain tightly focused

on this mechanism, particularly on better analytical bounds for the gain window, broader mapping of the geometry dependence, and controlled studies of whether limited attitude variation — informed by optimal steering approaches such as those described by Dachwald (2005) and McInnes (1999) — can delay the eccentricity and drag-driven breakdown without changing the fundamentally transient character of the effect.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Vitae



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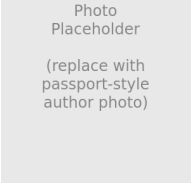
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